

Weekly Progress Report

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DATE & TIME	NO. OF HOURS	ACTIVITIES / TASKS	Supervisor Signature
May 6- 8	24 Hours	Day 1: Onboarding & Setup • Attended alignment meetings with the Machine Learning and QA/Annotation teams. • Set up local development environments and secured GitHub access. • Learned to navigate the repository using tools like Codex CLI. • Began annotating structural PDFs (marking beams and columns) to export as JSON ground-truth data. • Annotated a 75-page structural PDF set (AWS PHL 1b). • Configured local QA environments to test GitHub Pull Requests (PRs). • Ran benchmarking scripts to evaluate model accuracy on pending PRs. • Debugged and fixed PR51 by adjusting confidence thresholds and geometric boundaries to resolve false positives. • Attended a QA alignment meeting with the engineering team.	

		• Annotated the "Hope Landing" and "CFMC" structural PDF sets. • Ran evaluation workflows and discovered high volumes of false positives. • Manually traced errors backward through the detection pipeline. • Analyzed pdf_engine.py and the PDF Cluster Router for potential misclassifications. • Reviewed thresholding and deduplication phases to verify LightGBM model scoring.	
May 11 - 15	40 hours	• Annotated the Phil 102 structural PDF set and exported JSON ground-truth data. • Discovered a "zero detection" bug in the SteelSnap engine across four PDFs. • Investigated pipeline drop-offs in pdf_engine.py caused by rasterization or rendering failures. • Escalated a precise technical bug report to the engineering team. • Annotated the "Safeindust" PDF set and documented layout edge cases. • Reviewed PRs, identified potential model overfitting, and conducted Standard Deviation (STD) analysis on error metrics. • Performed Exploratory Data Analysis (EDA) using Pandas and	

		PyArrow on exported JSONs to map bounding box distributions. ● Generated and submitted an automated Python-based data science report to the supervisor. ● Completed ground-truth annotations for the "Tuscarora dc" PDF set. ● Participated in a live Project SteelSnap demo meeting. ● Observed real-time pipeline processing (ONNX Runtime, LightGBM) to evaluate bounding box accuracy and visual outputs. ● Tested the web interface via a Discord meeting and documented application crashes. ● Investigated backend pipeline bottlenecks where processing stalled between 70% and 80%. ● Formatted and submitted classification ticket CEN-637 to the Linear app. ● Reported replicated crashes and system stalls during the Project SteelSnap synchronization meeting. ● Participated in a Daily Stand-up Meeting (DSM) and Project SteelSnap alignment. ● Distributed 50 massive structural PDFs (100+ pages each) across the intern team to balance the workload.	
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May 18 - 22	40 Hours	• Attended the Daily Stand-up Meeting (DSM) to align on new detection testing batches. • Ran structural PDFs through the pipeline and documented detection behaviors to help the ML team adjust confidence thresholds. • Extracted detection outputs and formatted them into structured JSON files. • Performed Exploratory Data Analysis (EDA) on 10 PDFs using Pandas and PyArrow to map spatial segmentation and analyze dense clusters. • Participated in a machine learning progress meeting to review evaluation metrics and calibration goals. • Generated targeted ground-truth annotations for beam elements in bold, arc-heavy drawings to fine-tune the specialist_bold_r2 LightGBM model. • Ran pipeline batch testing on 32 PDFs using	

		run_sharded_benchmark.py. • Compiled an analysis report using Pandas to summarize model errors, false-positive spikes, and deduplication effectiveness. • Executed benchmarking workflows to validate outputs against expected ground truth across 35 structural PDFs. • Documented specific failure patterns, including False Positives, Missed Members, Missed Labels, and incomplete annotations. • Compiled a privacy-compliant analysis report detailing failure triggers based on source-data conditions. • Triaged a new batch of heavily vectorized PDFs to anticipate processing bottlenecks and ingestion strain on pdf_engine.py. • Conducted a secondary review of previously evaluated plans to verify error categorization and source-data anomalies. • Ran automated benchmarking (run_sharded_benchmark.py, compare_manual_counts.py) on 10 PDFs to quantify precision and recall degradation. • Investigated the PDF Cluster Router and discovered that excessive vector hatching caused	
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		incorrect routing to the global fallback model. - Analyzed deduplication failures using Shapely and isolated Column Geometric Baseline gating issues using diff_candidate_runs.py. - Authored a comprehensive technical report diagnosing pipeline hurdles and categorizing source-data errors. - Ran diagnostic scripts to prove that dense layouts inflated the feature space and degraded LightGBM specialist precision. - Attended a DSM to shift dataset prioritization toward company-provided structural PDFs. - Proposed pipeline architecture changes, including YOLO/ONNX masking for markups, PDF Cluster Router calibration, and updated spatial matrices.	
May 25 - 29	40 Hours	- Participated in a column-annotation coordination meeting and reviewed a 108-page structural PDF. - Analyzed Column Geometric Baseline failures, specifically False Positives, Missed Labels, and Missed Members. - Proposed geometric heuristics using Shapely (fill ratios) to filter out non-column squares. - Recommended integrating the YOLO	

		Symbol Detector and PyMuPDF to extract grid axes and infer missing labels. • Attended an evening DSM to evaluate the combined Geometric + ML approach for beam detection. • Validated four structural PDF plan sets and aligned on drawing interpretation guidelines. • Documented how source-data artifacts (e.g., heavy hatching) degraded the Geometric Signal-to-Noise Ratio (GSR). • Diagnosed "Cluster Bleed" in the PDF Cluster Router, where varying line weights misdirected drawings to the wrong ML specialist. • Reported edge cases where noisy linework caused geometric deduplication to incorrectly merge false positives into valid coordinates. • Filtered the PDF dataset to remove low-quality plan sets that were artificially deflating F1 evaluation scores. • Conducted targeted column detection testing and manually annotated complex column members. • Wrapped core detection modules (PyMuPDF, LightGBM, ONNX Runtime) into Docker	
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		containers to eliminate local environment errors. • Attended a DSM to align on a two-week client sprint, prioritizing column detection to anchor beam-to-column segmentation. • Investigated "High-Vector PDF" processing failures where pdf_engine.py crashed due to excessive vector loads. • Prototyped preprocessing strategies for dynamic layer flattening and vector simplification. • Proposed a raster-conversion fallback (circuit breaker) for pages exceeding safe vector count thresholds. • Researched PyMuPDF, Ghostscript, and TypeScript libraries (pdf.js) to handle backend flattening and frontend rasterization. • Standardized annotations using exact line-string geometries mapped to structural member centerlines. • Targeted noisy edge cases to prevent LightGBM candidate scoring models from learning false geometric patterns. • Manually linked semantic relationships between beam annotations and text callouts to train post-processing alignment.	
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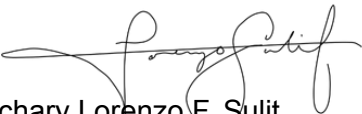
		Exported validated plan sets into standardized JSON schema to establish a new evaluation baseline via run_sharded_benchmark.py.	
June 1-5	40 Hours	- Conducted manual annotation tests on structural PDFs and ran the benchmark.py workflow. - Exported validated annotations into the structured JSON schema. - Loaded JSON files into Pandas to perform Exploratory Data Analysis (EDA), calculating standard deviations of coordinates and confidence scores. - Compiled statistical findings into a formal analysis report to identify systematic biases and presented it during a Project SteelSnap meeting. - Attended a synchronization meeting to define structural drawing use cases requiring immediate optimization. - Adjusted LightGBM model confidence thresholds using JSON data and standard deviation insights to handle specific use cases. - Annotated new PDFs and executed benchmarking scripts (benchmark.py,	

		run_sharded_benchmark.py) to evaluate precision and recall changes. • Helped clear an engineering bottleneck by using Linear and Codex CLI to review 10 Cross-Project Pull Requests (PRs) on GitHub. • Attended a strategic meeting (8:00 PM - 10:00 PM) aligning on the tight two-week client deadline for Project SteelSnap. • Executed dual-element manual annotations, mapping the geometric relationships between columns and beams within the same PDFs. • Ran updated JSON files through automated benchmarking workflows to evaluate False Positive and Missed Member rates in the LightGBM models. • Continued reviewing pending cross-project PRs using Codex CLI to maintain team momentum. • Participated in a strategic pause on active ML development to focus entirely on addressing data quality issues. • Manually annotated 20 structural PDFs explicitly targeting curved beam geometries to fix systematic detection failures. • Exported files to JSON and ran compare_manual_counts.	
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		py to establish a baseline for these edge cases. • Used Pandas to calculate the standard deviation between model predictions and manual ground truth for curved vectors, submitting a formal report. • Updated the local development environment with new repository modules and reviewed updated technical documentation. • Validated JSON coordinate accuracy for the newly mapped curved beam geometries against required spatial schemas. • Discovered a model performance trade-off during benchmarking: improved curved beam detection led to a spike in False Positives (over-extracting architectural markups). • Compiled findings into a use-case report and discussed per-source thresholding calibration strategies during a late-evening project meeting.	
June 8 - 9	16 hours	• Exported JSON ground-truth data for a batch of 50 structural PDFs. • Collated and organized previous technical reports, false-positive analyses, and edge-case documentation for	

		conversion into Markdown (.md) files. • Attended a major architectural alignment meeting (8:00 PM - 10:00 PM) introducing a new "Agentic AI" approach. • Handed over the collated reports to the tech lead to feed the new agent's knowledge base, allowing it to reason through complex edge cases. • Continued curating and formatting contextual data for the Agentic AI while traditional ML tuning remained paused. • Reviewed incoming PRs using Codex CLI, specifically debugging and validating a front-end UI fix where copy-pasted members snapped to the left instead of centering on the cursor. • Diagnosed a vulnerability where the agent's text-extraction layers failed to interpret unfamiliar typography from various external distributors, disrupting its reasoning.	
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